

DYNAMIC SYNTAX LABORATORY

DS Workbench

Complete technical walkthrough

Syntax as the growth of transparent semantic representations

The DynamicSyntax Python package and its DyLan foundation; the engine, grammar and application extensions built on it; Classical, Constructive and TTR interpretations; lexical assistance, corpus integration and verification; current limitations and a prioritized plan for broader coverage.

Checked against the working source on 4 October 2026.

ds-workbench.vercel.app

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Core walkthrough checked on 4 October 2026; research/relative-clause and community handover updates on 7 October 2026. Public application: <https://ds-workbench.vercel.app/>. This describes the implemented system, including the TTR referent-identity correction, live Greek lexical evidence, the controlled Jev comparison, and reusable finite-clause construction assistance. Section 2 documents the software foundation and additions; section 18 sets out the proposed next work. Future work is not counted as implemented coverage. Older design documents record earlier stages and can have outdated claims. Section 19 describes the local community handover work; it has not been published as a separate source/package release.

1 What the application is

DS Workbench is an interactive front end to an incremental Dynamic Syntax parser. It displays the semantic tree after each word, and can expose the lexical rules and individual instructions that built it. It has three semantic families: Classical DS, Constructive DS, and DS-TTR. Dictionary and corpus services enlarge the available vocabulary; optional language models suggest lexical analyses, finite-clause construction hypotheses or preferences. The DS engine constructs the derivation.

It is presently a research and teaching workbench with an expanding grammar fragment. It does **not yet meet the requested standard of reliably parsing an arbitrary Greek or English paragraph**. Running a paragraph through the interface and obtaining some complete sentences is different from covering open text. The application retains failed sentences and reports the distinction.

There are three independent questions when examining a result:

Question	What the application actually establishes
Does each word have an entry?	Lexical coverage for this input and selected vocabulary source.
Did a derivation complete?	The implemented grammar's tree requirements were satisfied, subject to its lexical and contextual assumptions and available semantic checks.
Is this the correct interpretation of grammatical language?	Completion alone does not establish this. Lexical sense, valency, reference and the grammar itself still require linguistic assessment.

The TTR bug discovered in this audit is a concrete illustration: the tree could complete while an incorrect variable collision corrupted the interpretation. Neither a green completion badge nor a type system proves the Python implementation correct.

2 Software foundation: dynamicsyntax, DyLan and our extensions

DS Workbench extends the dynamicsyntax Python package, a Python port of the Java **DyLan Dynamic Syntax parser**. The software lineage is Java DyLan, then the Python DynamicSyntax implementation, then the modified implementation and web application described here. Dynamic Syntax is the linguistic framework; DyLan and dynamicsyntax are software implementations of it. The system's basis is therefore a DS parser with substantial TTR machinery, rather than a generic TTR library to which a parser was subsequently added.

The repository's README credits the Java-to-Python translation to @incrementalizer. Its package meta-data names **Arash Ashrafzadeh** as author of the dynamicsyntax distribution and identifies the upstream repository as <https://github.com/incrementalizer/DynamicSyntax>. DyLan is linked there to the Dynamics of Language project. Those credits belong to the foundation; the workbench extensions do not imply that we originated the parser architecture or TTR.

Package name and implementation namespaces. Python users import dynamicsyntax; that package exposes the high-level parse() and icp() APIs, grammar discovery and result objects. Most engine implementation lives in the dylan namespace: tree and label classes, formulas, lexical/computational actions, parser search and the context DAG. These two namespaces are parts of the same source distribution, not

two independent parsing servers. The alias `ttr` resolves to the bundled `2015-english-ttr` grammar; it does not select every semantic implementation in the application.

What we retained and what we changed. The inherited engine already parsed incrementally, executed lexical actions, represented alternatives in a DAG, manipulated TTR formulas and supported backtracking. The repository also had GUI/web, visualization and induction work before the current workbench. Our contribution is an extension of that codebase, including changes to its core, rather than just a new display around an unchanged library.

Layer	Foundation retained	Workbench extensions and changes
Parser and context	Incremental parser, lexical alternatives, context DAG and repairs	Completion recovery, explicit limits, statistics, alternative analyses and replay/state isolation fixes
Trees and action language	Nodes, pointer, labels, IF/THEN/ELSE, make/go/put, MERGE and LINK machinery	Extended modal and feature checks, unfixed-node constraints, native semantic operations and operation-level playback
TTR semantics	Record types/functions, field dependencies, record composition and English resources	Branch-local freshening, preservation of participant constants, rejection of invalid manifests and tests of argument identity
Other semantic systems	Shared DS execution framework and formula interface	Classical lambda/choice terms and Constructive dependent terms, nominal typing, separate semantic profiles and Constructive Coq export
Grammars and vocabulary	Existing resource loader, action templates and legacy grammars	Native English/Greek grammar resources, reviewed clitic fragments, explicit construction families, WordNet/GDT candidates and validated model proposals
Application and sources	Public Python entry points and earlier interfaces	Current HTML/JavaScript/SVG workbench, isolated HTTP workers, sentence/paragraph/dialogue flows, OpenRouter, Jev controls, source evidence and deployment

How the additional semantics enter the engine. `semantics.json` declares a native grammar's back-end, axiom, constants, predicate signatures and nominal hierarchy. Native type/formula syntax is dispatched through the existing label and effect factories. Trees carry the semantic profile during cloning and search. Elimination then uses the appropriate native composition rather than TTR record composition. The `formula/mltt/` directory contains shared native term infrastructure used by both Classical and Constructive modes; its folder name does not mean Classical interpretation has become Constructive. Each mode has different permitted terms and interpretation policies. TTR remains on its own record-semantic path. Section 4 gives concrete resulting meanings.

For example, native `a man knows you` executes DS actions over the native grammar and composes a choice term or a dependent witness. It is not obtained by running the legacy TTR grammar and renaming fields afterward. Conversely, fixing a TTR referent collision repairs the inherited record path; it does not make the native grammar's new constructions available to TTR automatically.

Evidence from the repository history. Commit `57a6f57` on 25 September 2026 introduces the local graphical workbench; `4ea554a` adds the native Classical and Constructive semantic modes; `cac8923` restores the Classical common-noun restrictor structure. Later checkpoints include English LINK relatives (`f53a650`), WordNet/provider integration (`064e2fc`) and incremental lexical reconsideration (`84f28d8`). These identify development stages, not a complete attribution of every earlier change. The 4 October de-

ployment also contains subsequent local changes, including the source integration and TTR fix. 84f28d8 alone does not reproduce that deployment: its staged hash manifest and source snapshot are also needed. No new published PyPI release is claimed.

Installing `dynamicsyntax` from PyPI therefore does not by itself reproduce this workbench. The deployed application uses the modified source checkout and its grammar/data files. A reproducible workbench release needs a pinned source revision, resource hashes and dependency/runtime information, as proposed in section 18.

Linguistic sources and code licensing. The theoretical sources are distinct from the software implementation. The Constructive implementation is motivated by Chatzikyriakidis (2025), *Constructive Dynamic Syntax*, **Languages 10(11):269**, using modern type theory and nominal subtyping. Greek placement and PCC work draws on Chatzikyriakidis’s 2010 thesis, especially chapters 3 and 6, and the cited Cypriot/Pontic work. The historical panels describe evidence and a proposal about change; they are not executable grammars of Koine or medieval Greek. These are implemented fragments motivated by the cited analyses, not complete implementations of every construction discussed in those publications.

The inherited package metadata labelled the code BSD-3-Clause while `LICENSE` deferred to DyLan. Inspection on 7 October 2026 found LGPLv3 in the referenced DyLan repository’s `LICENSE.txt` (commit `ac70d021e5dc0eab24f185e5a34bbca19c0fc772`). The unsupported BSD metadata has been removed and the upstream licence text retained. This records the conflict; terms for all later contributions still need resolution before a public workbench release. See the [provenance notices](#). Code licensing does not put WordNet, GDT, corpora or dictionary content on the same terms; their separate conditions are discussed in section 12.

See the [native semantic design](#), the [constructive specification](#), and the [Greek source notes](#). Read their dated sections as development history; the coverage inventory below describes the current state.

3 What all three implementations share

A DS state is a tree plus a distinguished **pointer**. Addresses identify positions: `0` is the root, `00` its argument daughter, `01` its functor daughter. Labels include the node address `Tn`, a type `Ty`, a formula `Fo`, grammatical features, and requirements such as `?Ty(t)` or an unresolved formula.

The initial tree asks for a proposition. Each word selects a lexical program. Its conditions inspect the existing state; its instructions create nodes, move the pointer, add or remove labels, introduce formulas, or abort that candidate. Computational rules perform operations such as anticipation, completion, thinning satisfied requirements, substitution, MERGE and function application.

For a simple transitive clause, subject information occupies one branch; a verb creates or fills a function whose open object requirement is satisfied by the next NP. Composition proceeds up the tree. The stored alternatives allow the parser to revise an earlier lexical choice when later input makes it fail.

The context DAG records states and transitions, including word contributions, alternative paths and repairs. It is distinct from the semantic tree displayed at any one step. A new speaker can continue the same semantic tree; starting a new proposition introduces another tree in the context.

This shared machinery does **not** imply identical trees, identical grammars, or identical meanings across semantic systems. For example, Classical common nouns and Constructive common-noun types have different internal structures. The Classical and Constructive grammars are not translations of a completed TTR parse.

4 Classical, Constructive and TTR meanings

	Classical DS	Constructive DS	DS-TTR
Typical types	e, t, function types, a distinct common-noun restrictor type	object, human, nominal types, Prop, CN, functions, dependent Σ/Π	DS node types together with structured record types and record functions
Basic representation	Lambda terms, predicates, ε/τ choice terms and quantifier constructions	Typed terms, witnesses, projections and dependent types	Fields for individuals, events and constraints; dependency between fields
Indefinite	An ε term over a noun restrictor	A dependent pair/witness, closed existentially in the interpretation	An individual field constrained by a predicate field
Composition	Typed application and beta reduction	Typed application, nominal coercions/projections, beta reduction and witness closure	Record-function application, path resolution, freshening and asymmetric record merge
Current assisted vocabulary	Native English and Standard Modern Greek	Native English and Standard Modern Greek	Not implemented
Coq export	Not the Constructive exporter	Available for the supported term language	Not this export path

A concrete comparison. Enter a man knows you. with assistance off. The native implementations produce:

```

Classical:
  know( $\varepsilon$  x0:e. man(x0), you)

Constructive, raw:
   $\Sigma$  p00:( $\Sigma$  x:man. T). know( $\pi_1$ (p00), you)

Constructive, simplified:
   $\Sigma$  x:man. know(x, you)

```

The Constructive indefinite provides a witness p00. Its first projection supplies the individual to the verb. The final interpretation closes the witness. An interpretation type is not itself a constructed proof that the assertion is true.

After the freshening correction, the corresponding TTR record contains these fields, shown in a readable order without changing their names:

```

[ x0 : e
| p0 == man(x0) : t
| x1 == you : e
| e0 == know : es
| p2 == pres(e0) : t
| p5 == subj(e0, x0) : t
| p4 == obj(e0, x1) : t
| head == e0 : es ]

```

x0 supplies the subject and x1 the object. e0 supplies the knowing event; the remaining fields constrain its tense and participants. Display order and fresh field names are implementation details; the dependencies are essential. Different field names do not assert that the referents must denote distinct people: two fields may deliberately have the same manifest individual.

TTR is itself a dependent type-theoretic framework. Calling one menu option “Constructive” does not mean TTR is intrinsically nonconstructive. The labels identify different implemented semantic calculi.

Important native details. Classical determiner phrases use an explicit common-noun restrictor structure, not just a flat predicate pasted onto an entity. Constructive common nouns inhabit the grammar’s CN universe, with nominal inclusions used during application. Satisfying a pending tree type requirement is kept distinct from checking whether a function can consume a particular argument. Function domains require the appropriate variance rules.

Definites need contextual assumptions. Constructive output such as `definite_s1_010` names an assumed referent at a sentence/node position; it is not evidence that the app identified the uniquely intended door in the world. The interface displays the assumptions.

The existential-scope control supports a particular universal/indefinite configuration, for example Π doctor. Σ patient. ... versus Σ patient. Π doctor. It changes the supported closure policy; it does not enumerate every possible natural-language scope reading.

Finite complements have local witness closure. Constructive embedded contents use a Content representation exported as a Coq type. The attitude predicates remain opaque: there is no general logic of belief, factivity or possible worlds.

5 The TTR defect found and corrected

Previously `ttrput` passed the live dialogue Context to a freshening method that expected a Tree or parser tuple. The record implementation silently returned an unchanged clone for an unsupported argument. Both noun phrases therefore retained the field name `x`.

Later composition correctly followed the implemented asymmetric-merge operation, but the inputs incorrectly reused a label. One field overwrote the other. In a `man knows you`, `you` disappeared and both participant constraints referred to `x`. That is an implementation error, not a consequence required by TTR.

The correction allocates fresh names from **the actual branch receiving the formula**. Each cloned branch carries its own variable pools. Using the source tree held by the context would also be wrong during speculative search, because it would mutate the allocator shared by sibling alternatives. Unsupported freshening arguments now raise a descriptive error instead of doing nothing.

Regression tests inspect participant dependencies and retained individual constants in all three full bundled English TTR grammars. They cover two indefinites, name reversal, repeated names, speaker-dependent you, repair, repeated requests and agreement between the final playback channels. Browser checks inspect the displayed record and JSON export. These tests do not certify the entire TTR port or all legacy grammar resources.

Earlier smoke tests checked successful parsing and predicate presence; two other tests explicitly froze the corrupted record or its shared participant variable. Those expectations have been corrected. The new tests assert semantic relationships, including who fills each argument role.

The audit also found that re-reading a record with a capitalized participant constant such as `A` could discard its manifest value. Such constants now parse, while reserved metavariables retain their meaning. An unparseable manifest is rejected rather than converted into an unconstrained field. Unsupported `ttrput` formulas are rejected rather than installed as opaque semantics.

6 Walking through the interface

Start with **Sentence**, choose the semantic system and then its grammar. Use a small known example first. With a native grammar, select the grammar vocabulary alone to isolate the reviewed entries, or dictionary/corpus candidates to permit deterministic lexical expansion. Neither needs an LLM.

Press **Build derivation**. The results distinguish parser completion, lexical coverage and independent grammaticality evidence. A missing word means no usable lexical entry was available. A known word can still fail because its action conditions do not fit the current tree. Consuming every token can leave an incomplete tree with unresolved requirements.

The playback controls expose three views:

View	What it shows
Words	Recorded parser states at word boundaries.
Rules	The selected lexical/computational actions replayed through the engine.
Operations	Individual effects such as make, go, put, delete, MERGE or reduction where replay is available.

These are not three different parsing algorithms. The engine supplies the word states; replay provides a finer explanation of the chosen path. Some legacy transitions cannot be replayed instruction by instruction and are explicitly grouped. The word-boundary state remains authoritative. The trace is not a complete debugger recording every explored and abandoned search branch.

Click a node to inspect its full labels, formula and open requirements. Node types wrap to remain visible. Long formulas can be expanded in the tree; this only changes presentation and makes no model or parser request. Pan, zoom and Fit help with larger trees. Playback retains the pointer and changed-node cues.

The meaning panel follows the selected state. A partial root formula is not a complete interpretation. The Constructive panel can also display a simplified term. If more complete analyses were requested, the reading selector changes the tree, meaning and corresponding export together.

Export JSON preserves result data, lexical provenance, available trace and context. **Export Coq** emits the supported Constructive meaning and its declarations. Local tests can compile these files when `coqc` is installed; the public server does not run Coq. Compilation establishes type correctness of the exported representation, not a proof that the input sentence is true.

7 What the LLM does, and what works without it

Without an LLM, the application already has the parser, grammar programs, semantic compilers, reviewed lexical entries, bundled morphology, WordNet, Greek corpus hypotheses, source search, playback and exports. A dictionary candidate is still a hypothesis about which sense/frame fits the sentence. For example, WordNet can supply many senses of *enter*; it does not thereby choose the intended one.

With OpenRouter assistance, the sequence is:

The selected chat model can propose a lemma, word class, sense, nominal domain, argument frame, morphological features or a finite-clause relation. It receives context and, on a retry, information about the actual failed derivation. The compiler instantiates an allowed action template for the selected semantic backend. The model does not submit arbitrary Python, arbitrary DS programs, or an unchecked final tree for display as a verified parse.

The helper can request missing vocabulary before the first DS attempt as well as corrections after a failed attempt. Optional Greek source retrieval precedes the corresponding model proposal; it does not wait for a successful derivation.

Validation checks the permitted fields, word forms, domains, arity and template constraints. Predicate identifiers are controlled/generated. Protected function words cannot simply be replaced by a content-word guess. Existing alternatives remain available. A valid batch is installed and DS runs again.

The construction interface separates surface words from reusable DS operations. Twelve declared relations cover cause, temporal relations, conditions, exceptions, concession and contrast. Common connectives already work without AI. The model can prefer a contextual reading of ambiguous *since*, *while*, *as*,

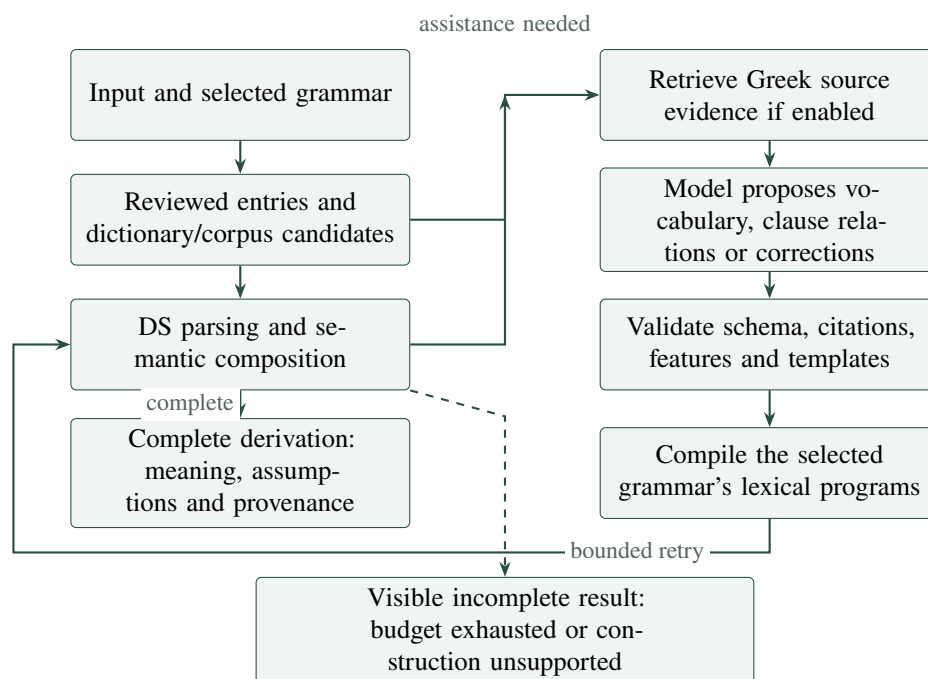


Figure 1: Missing vocabulary can trigger assistance before the first DS attempt; an incomplete derivation can trigger a correction. Optional Greek retrieval supplies evidence for the proposal. Compilation and DS execution still determine whether a derivation completes. Calls and retries are bounded.

once, ενώ or αφού, or propose a previously absent lexicalization such as conditional provided or causal καθότι. Missing-word requests accept both content-word and construction hypotheses, so an unfamiliar conjunction does not have to be forced into a noun/verb schema. A failed derivation can trigger a targeted construction request. The compiler validates requested surfaces, relation IDs, reviewed readings and signatures, then instantiates loaded templates. It does not execute generated code or add arbitrary types.

All original programs remain available; model preferences affect their order. The panel **How the clauses were connected** shows proposals, explanations, compiled programs and the relation on the actual DS path separately. A model preference may not be used. Known-word preferences reset between paragraph sentences; new lexicalizations remain request-local hypotheses and are marked as reused when appropriate. Repeated ambiguous surfaces within one sentence retain normal DS alternatives because occurrence-specific model ranking is not implemented. Jev remains the separate incremental lexical-sense selector; this construction interface uses the user's selected chat model and full context.

This explains both the usefulness and the limits of assistance. It can supply a missing lexical item, improve a valency hypothesis, or instantiate an existing construction with another connective. It cannot conjure an unimplemented relative construction, auxiliary system, reflexive-binding rule or PP attachment procedure. Those require grammar implementation. Even a compiled lexical proposal can select the wrong sense or frame while producing a type-compatible derivation.

Open-text mode makes at most three provider calls per request, shared across a paragraph, with bounded DS retries. The helper has a 48-second budget and a vocabulary call can use up to 22 seconds and a construction-only call up to 16 seconds within the remaining allowance. Combined requests count once. Rate limits such as HTTP 429 can stop further model calls; remaining usable entries can still be tried by DS. The UI reports actual model use and unresolved gaps.

This assistance can inspect the available full text. It is therefore not an experiment in strictly prefix-only online human language processing, even though DS builds its trees incrementally.

Connecting. Open Vocabulary & AI, enter an OpenRouter key, connect, choose a model, and choose the corresponding vocabulary mode. Connection checks the key and obtains compatible model information;

it does not itself parse with a model. The listed model inventory and prices come from the provider and can change. The selected model is used for the supported vocabulary requests.

The key is held in the tab’s JavaScript memory, not local storage or exported results. It goes to the DS backend for that request and onward to OpenRouter; the application does not persist it. Refresh or disconnect clears the tab’s copy. Inference is billed to the user’s provider account. Free application hosting does not make model calls free.

Assistance is implemented for native English and Standard Modern Greek. It is not implemented for TTR or generally for the other Greek varieties. Dialogue does not run the paragraph/sentence assisted retry pipeline.

8 Jev is a separate optional path

Jev is used as a typed preference service over lexical alternatives and parser state, not as the selected general chat model. The current provider pin is `typesafe/jev-1.13-20260917` on OpenRouter. Choosing another analysis model does not silently change that pin.

Its incremental lexical path uses the observed prefix, candidate information and current tree to prefer a sense. DS still checks frames and every proposed revision. Alternatives remain available for backtracking; a failed revision restores the previous path. Calls and reconsiderations are bounded, including an eight-fresh-request parse budget in the decision client.

A visible comparison. Open **See Jev in action**, connect OpenRouter, and try `John lends a book to Mary.` in English Classical or Constructive DS. The application executes two DS parses: one with normal lexical ordering and one with Jev preferences and reconsideration. The two result panels show the selected dictionary glosses, composed meanings, completion status and search statistics. Expand a decision to see the words observed, the entire probability distribution including uncertain, confidence, latency, available reported cost and any provider error. **Watch this revision** jumps to the corresponding backtrack in the actual DS action playback. JSON export retains the comparison.

The 4 October live check produced the following result in both native backends:

Point in the example	Observed behaviour
DS without Jev	Completed with the dictionary sense of <code>lend</code> meaning “bestow a quality on”
Jev at <code>John lends</code>	Chose uncertain; neither object nor recipient had been supplied
Jev after <code>John lends a book</code>	Preferred “give temporarily”; bounded DS replay found no compatible continuation yet and retained the current tree
Jev after <code>John lends a book to Mary</code>	DS replay accepted the preferred sense; the final formula used the temporary-transfer predicate with <code>book</code> and <code>Mary</code> arguments

Both runs completed, so completion alone would have missed the lexical problem. The ordinary reading of the sentence supports the temporary-transfer choice; this is one inspected demonstration, not a measured general improvement. There were three fresh Jev calls per backend. Outcomes can differ on later requests. The crane example permits reconsideration after later context, while the bank example deliberately leaves the sense underdetermined. The interface also reports unchanged choices, unavailable answers and incomplete comparisons. Model probabilities are preferences, not calibrated linguistic accuracy.

A later check on the deployed site again changed the final sense in both backends. Classical Jev remained uncertain after `a book` on that request; both accepted the revision after `to Mary`. The recorded public result retains the actual distributions and replay outcomes. The interface displays each request’s result rather than fixed demonstration answers.

How the comparison is controlled. `jev_comparison.py` prepares candidates once, including any permitted lexical-model fallback, and copies their source programs, metadata and semantic profile into two fresh parsers. It verifies a content hash of each installed inventory and resets the pooled metavariable bindings

between runs. Both receive the same grammar, initial context, scope and entry limit. The baseline does not call Jev. This matters because ordinary dictionary mode consults the bundled extension before WordNet, whereas Jev vocabulary mode consults WordNet first: switching those modes would change more than the model preference. The comparison therefore uses the Jev-mode inventory in both runs. It is not an off/on comparison of two different vocabulary sources.

The request option is `compare_jev: true` with `lexical_mode: "jev"`, normal search ordering and one requested analysis. It currently supports native English sentences only, not Greek, TTR, paragraphs or dialogue. The baseline has a four-second search allowance and runs without the animation replay overhead; the comparison has a 50-second internal budget after initial preparation starts. Its timings are consequently not a speed benchmark. If either run fails or times out, the comparison is inconclusive. The primary displayed tree is the Jev run; the baseline includes its selected entries, formula and derivation action names. The request cannot supply arbitrary lexical programs through this internal inventory-copying interface.

Lexical preferences and **search ordering** are separate controls. Search-order use has calibration gates tied to the grammar, code, provider/model and state representation. A configured key is not evidence that ordering is enabled or beneficial. Existing experiments did not demonstrate a backtracking benefit sufficient to enable those gates. The normal DS ordering remains meaningful and available without Jev.

The distinctions are thus: DS determines whether a path completes; the chat model proposes vocabulary and finite-clause relations; Jev can influence lexical-sense preferences among candidates. Neither service is an independent judge of natural-language grammaticality.

9 PP arguments, LINK modifiers and similes

Your distinction between arguments and LINK modifiers matters. The implementation has separate cases rather than one generic “preposition” analysis:

Use	Structural treatment	Example
Selected PP argument	A lexical frame opens a required slot; the selected preposition fills that slot	relies on Mary; a reviewed goal frame for Greek μῆκε στην πίστα
Optional VP/clause modifier	A LINK structure grows and combines its contribution with the host	Postverbal locative/manner/path modifiers in the supported fragment
Copular PP predicate	The PP supplies the predicate of the subject	a man is in the room
Nominal comparison	A LINK contribution uses a noun description as a comparison standard	like a wild peacock / σαν άγριο παγώνι

A selected argument contributes to the verb’s valency. Removing it may leave the selected frame incomplete. A modifier adds information to an otherwise available predication. The distinction is encoded by lexical frames and action templates; it is not reliably inferred just from the surface preposition.

The English motion example can compose a path modifier and then a comparison:

```
George walks onto the stage like a wild peacock.

manner_like(Σ x:peacock. wild(x),
            onto_goal(definite_01L10, walk(george)))
```

The Greek example uses its selected goal frame:

```
Ο Γιώργος μῆκε στην πίστα σαν άγριο παγώνι.
```

```
manner_like( $\Sigma$  x:peacock. wild(x),
            enter_goal(giorgos, definite_010))
```

Here the Σ expression supplied to the comparison operator is a **description type**, not a claim that a particular wild peacock exists in the discourse. The construction also does not assert that George is a peacock. Greek singular case/gender constraints are checked by the supported nominal templates.

The semantics of `manner_like`, `path` and `location` operators is deliberately limited. There is no complete event/spatial theory behind them. General NP attachment, all instrumental/comitative distinctions, arbitrary PP chains, referential or plural comparison standards and clausal comparisons remain outside the implemented inventory. English and Greek have explicit related implementations; they are not obtained by merely translating words.

10 Current construction coverage

Native English. The maintained fragment includes names and pronouns; indefinites, definites and universals; distributive `all` with supported plural nouns; nominal, adjectival and locative copulas; degree expressions; intersective attributive adjectives; active intransitive, transitive and selected ditransitive/PP frames; do-support negation; positive modals with supported bare verbs; clause and VP coordination; LINK adverb/PP modifiers; finite complements and causal clauses; and restrictive relatives with supported subject or direct object gaps. Relatives use LINK and an unfixed head copy, with MERGE and nominal refinement in the Constructive implementation.

Reusable finite clauses. A shared, parameterized construction now covers English when, whenever, while, before, after, since, until, if, unless, although, though, whereas, once and as, and Standard Greek `όταν`, `όποτε`, `ενώ`, `αφού`, `αν/εάν`, `μολονότι`, `παρότι`, `ωστόσο/ώσπου`. Ambiguous connectives retain distinct readings. These entries require finite clauses; they do not supply interrogative/relative when, nominal after, gerund complements, or multiword markers. A postposed clause uses LINK; a preposed clause requires a comma and another clause. For example:

```
When George opened the door, Mary entered the room.
John walks although Mary walks later.
If Bill shouts, John walks because Mary walks.
```

Both native backends parse this three-sentence paragraph without an LLM. The last sentence composes `condition(because(walk(john), walk(mary)), shout(bill))`. A separate three-sentence Greek check with `όταν`, `παρότι` and `αν` also completes in both backends. These are hand-written construction checks, not a new held-out accuracy measurement.

All relations order their arguments (`main`, `subordinate`) and preserve local content closure. They are explicit, opaque predicates. Temporal precedence, overlap, conditions and concessions are distinct from cause, but full event semantics, temporal anchoring, counterfactuals and discourse inference remain unimplemented. In particular, `condition` is not automatically material implication. Constructive Coq exports `declare Type -> Type -> Prop`; Classical uses its own proposition and choice-term semantics. No TTR translation is used.

Causal instance. English `because` and Standard Greek `επειδή` / `διότι` now connect finite clauses in the two native backends. Both `John walks because Mary walks later.` and `Because Mary walks later, John walks today.` run without a model. The preposed construction requires a comma and a following main clause. Postposed reasons grow a LINK. Preposed reasons occupy a content argument of a curried causal operator; that slot belongs to the connective, not to the verb's lexical valency. This is the implemented analysis, not a claim that every DS account must use that representation.

The reported sentence `George opened the door because he came later` completes with WordNet vocabulary and no model request, in both native backends. It also completed in live Jev mode, which retained the same lexical choices in the check. The Constructive result has the form `because(open(george, definite_door), later(come(george)))`, with the actual selected WordNet predicate identifiers retained in the export. Classical instead uses its definite choice term for the door. The antecedent choice for `he` is the existing most-recent-compatible-name heuristic; that assumption is now displayed.

`because(main, reason)` is an opaque relation between two locally closed clause contents. It is not reduced to conjunction, and does not supply causal inference or truth axioms. Constructive witnesses close separately inside each content; its Coq declaration is `because : Type -> Type -> Prop`. Classical retains its own `t` contents and choice terms. Missing reasons, missing main clauses and unlicensed `because` of phrases remain incomplete. Supported nested reasons and finite-complement attachments can produce separate readings. Coordination and negation do not have a complete causal-scope calculus.

Temporal words include `later`, `earlier`, `late`, `early`, `today`, `yesterday`, `tomorrow`, `soon` and their maintained Greek counterparts. Each contributes a distinct operator. Comparative/reference times and calendar anchoring remain contextual; a past surface form still does not provide a full tense/aspect interpretation. The causal addition is tested at the level of meaning as well as completion: following adverbs/PPs cannot erase the reason, and a modifier in a preposed reason must propagate before the comma opens the main clause.

The dialogue work adds `do/does/did` polar questions, local subject-bound reflexives and `Yes/No` answers to an immediately preceding complete question. This does not provide a complete English auxiliary, tense or question grammar.

Standard Modern Greek. The maintained native grammars combine a clitic fragment with copulas, degree predication, singular case-bearing common-noun DPs/adjectives, corpus-supported ordinary finite verbs, destination PPs and nominal comparisons. The clitic work includes accusative/genitive forms, person-sensitive restrictions, monotransitives/ditransitives, imperative clusters and strong PCC constraints. Negation, $\nu\alpha$, future particles and gerunds have explicit entries. Case-marked names permit supported SVO/VSO/OVS patterns, basic clitic coreference/doubling and restricted initial topics.

For `της τον έδωσε`, the corpus expansion can propose a ditransitive entry; the grammar links the accusative to the theme and genitive to the recipient, giving `give(pro, him, her)` under its context assumptions. A corpus observation of a transitive use of the same lemma is not enough by itself to infer this three-place frame. The builder records separate form and frame evidence where it combines observations.

Greek varieties. Cypriot, Pontic and Grico are smaller, separately reviewed fragments available in the native semantic families. They model selected placement environments and restrictions, not general dialect understanding. Cypriot supports a bare-clause/trigger contrast; Pontic uses the reviewed postverbal conditions; maintained Grico work includes selected cluster order and PCC behaviour. The interface's source and environment qualifications are part of the analysis. A sentence outside the reviewed inventory cannot be treated as evidence against the dialect's grammar.

Legacy TTR English. The bundled resources retain their own lexicon and construction inventory. Native English additions are not automatically added to these files. Some legacy entries fail non-strict loader validation and are skipped with warnings. A row visible in a lexicon file therefore does not prove the word/action is usable. The small `2026-english-ttr-test` resource is a test fragment, not a fourth broad English grammar.

Major gaps across the system. General agreement and tense/aspect, productive passive/auxiliary combinations, arbitrary `wh`-extraction, control and raising, unrestricted possessives, mass/collective readings and many plural constructions are incomplete. English relatives are not a complete theory of relativisation or islands. Greek does not have a general morphological analyzer, complete clause embedding or general `nov` relatives. Broad anaphora, discourse accessibility, presupposition and pragmatics remain unfinished.

11 Paragraphs and dialogue

Paragraph mode splits text into sentence spans while retaining the original text/offsets. The splitter accounts for supported punctuation, abbreviations and numerical cases; it is still an engineering tokenizer. Each sentence is processed and gets an explicit result. Completed sentences contribute context; a failed sentence's parser changes and provisional lexical/context state are rolled back. Later successes are shown with context gaps rather than presented as a complete interpretation of the paragraph.

Basic named antecedents can support later pronouns. This is not a complete discourse-representation or reference-resolution system. Paragraph mode uses one selected analysis per sentence, not a global search through every possible combination of sentence readings.

Dialogue turns carry speaker/addressee metadata. **Continue tree** lets the next speaker finish the current structure; **New tree** introduces another proposition while retaining earlier trees for inspection. **sorry** invokes a bounded local repair of an earlier lexical contribution. Original words remain visible in the history.

For the requested example:

A: Did you burn
B: myself?
A: No

the native English analysis resolves A's you to B. B's myself must match the local subject B. The question content is `burn(speaker_B, speaker_B)` and the negative answer is `¬burn(speaker_B, speaker_B)`. The question is retained and labelled as question content rather than an asserted fact. This works without an LLM. General ellipsis, clarification, grounding and arbitrary repairs are not implied by this example.

Current input bounds:

Input	Limits
Sentence	500 characters; 40 tokens
Paragraph	6,000 characters; 400 tokens; 24 sentences; a sentence within it at most 2,000 characters/160 tokens
Dialogue	1–2 speakers; 12 turns; 80 tokens; 1,200 total characters; 500 characters per turn
Alternative complete analyses	Requested sentence limit 1–8, within bounded search

Time limits also matter. Dictionary-only paragraphs normally have a 23-second overall budget and a three-second sentence allowance. Assisted paragraphs can use a larger overall allowance within the hosted worker limit. A limit-hit result describes a bounded computation, not a linguistic impossibility.

12 Corpus and dictionary sources

Greek search connects to Svarna's existing remote service, with its available databases, corpus/variety, register and mode filters. Word/phrase searches use its word index; they are not a general Greek lemmatizer. Dialect examples stay associated with their source. Triantafyllidis lookup retrieves dictionary headword or entry results; a missing entry for a dialect form is not a claim that the word is invalid.

English search uses locally bundled, read-only SQLite indexes: Brown's 500 documents and 18 books from NLTK Gutenberg. This is not a live search of all Project Gutenberg or the web. Brown tags are removed from displayed text; Gutenberg segmentation retains source spans. WordNet supplies dictionary senses and mapped lexical-frame candidates. Query length, pagination and regex work are capped.

Selecting an attested example lets the user send the text or an explicit excerpt to the parser. Editing the analysis input does not rewrite the source quotation. Provenance and an edited-source indicator accompany exports.

The automatically compiled Greek vocabulary comes from a bundled **GDT training-split** export, currently 2,021 forms and 2,330 analyses. The held-out GDT test material is kept separate from that builder. Live Svarna and Triantafyllidis results also support investigation and, optionally, lexical proposals during parsing.

For Standard Greek Classical or Constructive sentences and paragraphs, enable **Use Svarna + Triantafyllidis during parsing** under Vocabulary & AI, with Open text assistance and a connected model. When a lexical model request is needed, the backend retrieves source observations before asking for proposals. If the existing entries suffice, no model request or automatic source lookup is needed. The switch starts unchecked and does not enable TTR or dialect assistance.

`greek_lexical_evidence.py` implements this request-local layer. It considers at most four requested content forms across the whole request, with up to two candidate dictionary lemmas per form. Known form/lemma mappings come from the GDT training index; otherwise an existing lexical hypothesis or a surface-form headword search supplies the query, with that uncertainty recorded. This is not a general morphological analyzer. It searches Svarna for the surface form, and Triantafyllidis for the candidate headwords. Only individual words/lemmas and the corpus filter go to these services; the selected model receives the excerpts together with the input text and parser context.

Svarna retrieval uses a selected small corpus: **UD Greek GUD** by default, or **UD Greek GDT**. These filters returned examples during integration checks; the Parliament filter returned zero even for a common-word probe, and a Europarl probe timed out. Neither is offered in this first runtime integration. Svarna does not expose original train/test split or sentence IDs in these results. Consequently, live GDT/GUD retrieval must not be counted as unseen evaluation on those corpora. This feature does not revise the earlier frozen coverage numbers.

The collector runs bounded concurrent HTTP reads, sharing an eight-second retrieval budget across model retries and paragraph sentences. It caches public observations, misses and source failures for that request only. Each lookup keeps at most two examples or dictionary entries, with explicitly shortened excerpts. Late results are discarded. A timeout, malformed response or failed filter check is recorded as unavailable, separately from a successful zero-result search. Further forms beyond the four-form limit are named as skipped. The model can still propose an explicitly uncited hypothesis; only DS determines completion.

The provider schema adds `evidence_ids`. The server validates every cited ID against observations supplied for that surface form before installing any candidate. A fabricated or cross-word citation rejects the proposal batch. The compiler and lexical action metadata retain validated references; paragraph and sentence JSON exports include the source text, query, retrieval time, entry ID or content fingerprint, source URL and available corpus metadata. A content fingerprint is not represented as an upstream sentence ID. Source text is data, never executable instructions, and the browser renders it as text.

Expand **Why this analysis?** to compare each word's dictionary queries, retrieved observations, model-inferred frame and citations. These are lexical candidates, not necessarily the entries used on the final DS path. ID validation does not verify citation relevance or establish the intended sense, exhaustive valency or grammaticality. PP argument/modifier distinctions still depend on explicit templates and grammar actions. There is no automatic grammar training, reviewed-lexicon promotion, dictionary mirror or Svarna-side API change.

WordNet and GDT frames are lexical hypotheses, not exhaustive valency descriptions. They can produce alternatives irrelevant to the present context. Code licensing and data licensing are separate; the GDT export carries its CC BY-NC-SA 3.0 attribution/license, and bundled corpora/dictionaries retain their own source conditions.

13 Frontend implementation

The current public interface is plain HTML, CSS and JavaScript, with SVG trees. There is no React application or browser Python runtime in this deployment. `workbench/app.js` owns application state: configuration, result, input mode, playback channel/index, selected sentence/reading/node, timer, viewport and expanded formulas. `openrouter.js` owns the key/model connection controls; `research.js` handles corpus and dictionary interactions.

At startup the browser requests `/api/config`. Parsing posts to `/api/parse/stream`, reads newline-delimited JSON from the response stream, buffers partial lines, and updates the available frames. Events include initialization, frames, lexical information, paragraph sentence results and the final result. The browser has a 60-second request abort boundary.

Tree layout measures text and reserves space for wrapped types and expanded formulas. The inspector uses the full stored labels. Expanding one formula changes the SVG layout rather than recomputing its meaning. The selected reading/state also drives copying and exports so they refer to the visible analysis.

The playback banner separates the final parse result from the current animation step. A completed parse can still be displaying an early, incomplete tree. **Show final result** pauses playback and moves to the final recorded state; an incomplete input retains its failure status. In paragraph mode the banner describes the selected sentence, not the completeness of the whole paragraph.

Large paragraph traces are bounded: a sentence result above the trace budget can retain only initial/final states, with an explicit note. This protects response size; it is not evidence that intermediate derivation steps did not exist.

The repository also contains an older Pyodide frontend under `web/`, a Flet GUI path, and LaTeX/Manim tooling. They are separate interfaces, not components of the public browser's runtime.

14 Backend and hosting

The hosted entry point is a FastAPI application, using the ASGI adapter. A local standard-library HTTP server is also available for development. Both call the same workbench parser API. The central routes include configuration, ordinary JSON parsing, streamed parsing, OpenRouter connection and research lookups.

A parse runs in a separate Python subprocess, with JSON input and JSON/NDJSON output. This matters because the inherited engine has pooled metavariables and other mutable parser state. Process isolation prevents separate users' parses from sharing those pools. The worker builds the selected grammar and semantic profile, installs permitted lexical candidates, parses, and serializes results.

The public adapter bounds request size at 32 KiB and response size at 4 MiB. It allows two parse workers and four lookup slots per instance. Parsing has a 55-second worker limit; the Vercel function has a 60-second configured limit. Timeout/disconnect handling kills, drains and reaps workers. These per-instance limits are not a claim about unlimited global capacity.

Hosted requests use the supplied key for the request; host credentials and arbitrary custom endpoints are not exposed as public parsing features. There is no app account database, persistent personal parse history or saved model-key store. Users preserve results by exporting them.

DS Workbench is hosted on **Vercel**, with the Python 3.12 runtime and a Frankfurt region setting. It does not require its own Azure container. Svarna remains an independent external service hosted at the Azure URL you supplied. Hosting the DS app elsewhere does not move or take ownership of Svarna.

Deployment uses `scripts/prepare_vercel.py` to copy the runtime, grammar files, frontend, selected evidence and read-only indexes into a controlled staging directory. A hash manifest records the inputs. The 4 October construction-family build contains 350 files, about 80.34 MiB, without local `.env` files or user keys. Vercel CLI publishes this staged snapshot; it is not automatic deployment of arbitrary files from the working tree.

Free-tier hosting remains subject to the provider's current quotas and terms; there is no fixed Azure-container charge for the DS app. Model billing is separate. Heavy concurrent traffic or larger corpora may require a different worker/storage arrangement.

15 Programming details that matter for correctness

Branch state and metavariables. A parser branch needs its own tree and variable-allocation state. Metavariable bindings have to be saved/restored around probes, failed candidates and read-only snapshots. Ordinary tree clones can share label objects, so trace capture uses deeper snapshots where mutable bindings would otherwise change an earlier displayed state. The TTR fix belongs to this class of issue.

The action DSL. Grammar entries are programs with IF/THEN/ELSE conditions, not labels assigned by a tagger. Template argument count and control flow are significant. A loaded word can fail when a precondition is false. Non-strict legacy loading may skip malformed entries; strict loading and grammar linting are useful developer tools.

Unfixed nodes and MERGE. An unfixed node represents information whose exact attachment is pending. Merge checks formula and type compatibility together with relevant case/person/number/gender constraints. Local-unfixed positions are not interchangeable with arbitrary descendants or LINK branches. Modal box and diamond tests also differ: checking all reachable nodes is not the same as finding one suitable node, and the box can be vacuously true.

Typed native composition. The immutable term AST performs capture-avoiding substitution and beta normalization. Constructive coercions can project from a Σ refinement to its nominal head. Quantified objects and the newer auxiliary combinators require recognizing which daughter supplies the consuming function; always assuming the right daughter would be incorrect.

Fresh TTR fields versus intentional sharing. Independently introduced records need fresh labels. Intentional coreference and head dependencies need to survive. Making every merge rename every field would break legitimate sharing; the repair belongs at introduction, not in a blanket change to ++.

Bounded search. Lexical alternatives, computational adjustment and completion are bounded to prevent runaway requests. Native workbench parsing retains all lexical entries by default, whereas the library/TTR default retains up to three per surface form unless configured otherwise. End-of-input native completion can try alternative paths, including a bounded 32-candidate primary recovery. Requesting multiple analyses uses additional bounded enumeration. These bounds mean failure can result from search exhaustion as well as missing grammar.

Generators and maintained resources. Common English/Greek extensions are generated into marked grammar sections. `build_everyday_extensions.py` invokes the PP and dialogue extension builders. Editing only a generated lexicon tail can lose the change on regeneration. The older fragment generator is not a safe blanket replacement for subsequently maintained grammar cores. Changes need review in both the generator and the emitted programs.

Reproducible lexical evidence. WordNet/GDT builders use pinned inputs and stable identifiers. GDT occurrence frames and inferred combinations retain their provenance separately. Vocabulary validation is atomic for a proposal batch; a half-installed invalid proposal must not leave a parser in a mixed state. Paragraph rollback also needs to restore the semantic profile and lexicon, not merely the visible tree.

Induction is separate. The repository contains offline TTR-oriented grammar induction/EM machinery, dataset splits and candidate-lexicon tooling. The public workbench does not run that training pipeline when a user submits a sentence, and a chat-model proposal is not the same operation as formal grammar induction.

16 Coverage evidence and what it does not show

The UI exposes two frozen engineering measurements:

Sample	English, each native backend	Greek, each native backend
1 October unseen-source baseline	0/6 paragraphs; 0/19 sentences complete	0/6 passages; 0/18 sentences complete
Six hand-written assisted development passages	2/3 passages; 6/9 sentences complete	1/3 passages; 5/9 sentences complete

The Greek baseline used groups of three consecutive test-set sentences because original paragraph boundaries were unavailable. Neither small sample is a representative accuracy estimate. The baseline predates subsequent grammar extensions, so it is also not a fresh evaluation of every change described here. The development passages are not an independent unseen benchmark.

Regression tests answer another question: whether implemented behaviour and important exclusions remain intact. Thousands of passing assertions do not mean thousands of arbitrary sentences are covered. Current tests include positive and negative grammar cases, native typing/Coq checks, parser replay, corpus provenance, assistance validation and server behaviour. The new TTR tests add explicit semantic identity checks.

The TTR correction checkpoint passed **1,636 tests**. The subsequent 4 October full run including live-source integration passed **1,655 tests, with 3 skipped and 1 deselected**. After retaining complete model explanations, a focused run passed **90 tests**; the evidence-specific suite contains 20 passing cases. These are dated runs, not a new test execution for this report revision. The deselected full-suite case is the existing external LaTeX compilation smoke test described below. Browser checks verified corrected TTR fields, displayed meaning, source quotations, citations, JSON export and key privacy.

After adding the controlled Jev comparison, the full suite passed **1,674 tests, with 3 skipped and 1 deselected** on 4 October. The comparison and existing lexical-selector tests passed 41 focused cases. Live browser checks exercised both native backends, equal inventories, probability displays, revision playback, JSON exports, key privacy, mobile layout and returning to dictionary-only mode. The full-suite exclusion remains the same external LaTeX smoke test; the current technical report is compiled separately with XeLaTeX.

The subsequent causal-clause checkpoint passed **1,733 tests, with 3 skipped and the same 1 deselected**. The 59 causal-specific cases also passed after the final backend-consistency guards. They check complete meanings, negative cases, local scope, modifier preservation, Coq output, replay and paragraphs. Live dictionary and Jev checks both accepted the user's reported sentence; these are construction/integration checks rather than a new coverage estimate.

The reusable construction-family tests additionally check temporal, conditional, concessive and contrastive clauses; nested meaning preservation; model-selected readings and new lexicalizations; rejection, request budgets and paragraph preference reset. Real OpenRouter checks completed provided and *καθότι* examples in both native backends. In the deliberately ambiguous since case, the model preferred a causal reading; no uniquely correct reading is claimed. Two hand-written three-sentence paragraphs, one per language, completed without AI in both backends. These focused checks do not revise the held-out baseline.

At this checkpoint, 1,809 distinct regression cases have passing results across the full and focused runs. The full run passed 1,806 cases with 3 skipped and 1 external LaTeX smoke test deselected; its sole failure was a diagnostic-text assertion. That wording was restored. The final focused run passed all 171 plural/construction/causal cases, including that assertion and two newly added combined-request checks. Live browser verification covers the construction panel, actual DS choice, JSON export, key privacy and mobile layout.

Live source integration was also checked on the deployed site: Classical Η ερευνήτρια περίμενε. completed, and the Constructive paragraph Η ερευνήτρια περίμενε. Η Μαρία είναι εδώ. completed both sentences. The retrieved Svarna examples of περίμενε included imperatives; indicative past morphology in the input remained an inference. This usefully exposes the boundary between an observed form and an interpreted construction. It is an integration check, not a new coverage benchmark.

The remaining evaluation task is a new frozen, genuinely unseen bilingual sample, with completion, correct sense/frame/reference, semantic adequacy, runtime and provider cost reported separately. The wide-coverage objective remains open until that evidence supports it.

17 Where to read and change the code

Responsibility	Main files
Public Python API	src/dynamicsyntax/_parse.py, parse_result.py
Parser search	interactive_context_parser.py, dag_parser.py
Tree and context	tree.py, word_level_context_dag.py
Lexicon/action loading	lexicon.py, effect_factory.py
Native terms and semantics	terms.py, semantics.py, semantic_effects.py
TTR records/freshening	ttr_record_type.py, ttr_formula.py, ttr_fresh_put.py
Native dialogue semantics	native_dialogue.py
Finite-clause construction family	clause_inventory.py, causal_effects.py, build_causal_extensions.py
Model construction compiler and provenance	construction_assistance.py, assisted_parsing.py
Request orchestration	workbench_api.py, paragraph_workbench.py, dialogue_workbench.py
Lexical assistance	assisted_parsing.py, lexical_expansion.py, lexical_provider.py
Jev preferences and controlled comparison	lexical_selection.py, jev_comparison.py, decision/
Research sources	research_sources.py, english_sources.py
Greek evidence during parsing	greek_lexical_evidence.py, greek_lexical_dictionary.py
Hosted/local HTTP	workbench_asgi.py, workbench_server.py
Browser	workbench/index.html, app.js, styles.css, openrouter.js, research.js
Grammar resources	src/dynamicsyntax/grammars/, including per-native-grammar semantics.json
Expansion generators	build_everyday_extensions.py, build_pp_extensions.py, build_dialogue_extensions.py
Deployment	prepare_vercel.py, deploy/vercel/

To run the local UI with the existing environment:

```
.venv/bin/python -m dylan.workbench_server --port 8769
```

To check the referent-identity regression:

```
.venv/bin/python -m pytest -q tests/test_ttr_referent_identity.py
```

The application test suite can be run with the legacy external-Latex smoke test excluded when its dsttr/rtrees dependencies are unavailable:

```
.venv/bin/python -m pytest -q -k 'not test_compile_tex_smoke_if_latexmk_available'
```

That exclusion concerns the separate legacy LaTeX renderer. It does not remove the TTR parsing or semantic-identity tests. Provider/browser checks are separate scripts; model calls require an explicitly configured account and may incur cost.

18 Forward plan: from fragments to broader verified coverage

Research update, 7 October 2026. The [theoretical coverage review](#) and its PDF now classify the literature into eighteen categories and ninety-one tracked topics. These are research topics, not a claim of ninety-one implemented analyses. The accompanying JSON catalogue records evidence level, source passages, current implementation, backend obligations and priorities. Its implementation programme refines the earlier empirical roadmap below.

The first resulting increment adds native English nonrestrictive relatives: *John, who Mary knows, walks.* yields *walk(john) ∧ know(mary, john)*. The grammar uses LINK, a shared unfixed head copy, MERGE and separate propositional evaluation, following Cann, Kempson and Marten’s §3.1 analysis. Classical and Constructive meanings retain their own witness/term handling. Named matrix heads and local subject/object gaps are supported; Greek *πov*, quantified hosts, embedded supplement projection and TTR parity remain separate work. See the [implementation note](#).

The next objective is to parse substantially more unseen English and Standard Modern Greek text with inspectable DS derivations and defensible meanings. Increasing the number of green completion badges is insufficient: an improvement must preserve argument roles, reference, quantification and known exclusions. The following priorities are proposed work, not shipped features or promised accuracy. The first deliverable should combine a fresh failure inventory with productive lexical morphology, followed by the construction families that the inventory shows to be most valuable. TTR correctness remains an immediate concern throughout, with its own explicit expansion track.

Priority 1: establish a current, reproducible coverage baseline. Freeze a new bilingual collection with source/document IDs, genre, sentence boundaries and document-level development/test separation. An initial engineering target is 50 development paragraphs and 50 held-out paragraphs per language, across at least three registers. This is a proposed collection size, not an existing dataset or evidence of population-level accuracy. Keep hand-written diagnostic pairs separate from natural-text measurements. Freeze expected semantic roles and readings on a reviewed subset and record ambiguity where appropriate.

Run the same items through grammar-only, dictionary/corpus, model-assisted and model-plus-live-evidence configurations where supported. Compare native backends separately and evaluate TTR against its own available grammar. Report lexical coverage, complete sentences, complete paragraphs, reviewed semantic adequacy, overgeneration on diagnostic negatives, latency, calls and cost. Classify each failure as a missing lexical form, unsupported construction, conflicting features, wrong lexical hypothesis, search limit or unresolved discourse. Keep every input in its denominator. Once a test item guides a fix, treat it as development material and replace it in the untouched evaluation set.

Priority 2: make vocabulary productive through morphology and better evidence. Add a morphology/lemma interface that returns competing analyses with provenance, rather than requiring every inflected spelling to be hand-entered. Greek needs case, gender, number, person, tense/aspect and voice distinctions; English needs reliable inflections, irregular forms and auxiliary compatibility. Recognizing a feature does not yet implement its DS construction or tense meaning. Choose resources with usable licenses and documented training/evaluation overlap; model or tagger output must remain labeled as predicted analysis.

Use those analyses to query dictionary lemmas and select relevant sense passages. The current collector takes short entry prefixes and consults at most four forms; it cannot claim to have considered an entire dictionary entry or paragraph. Prioritize unresolved forms, retrieve relevant spans with their surrounding context, and record which ambiguity the evidence addresses. In particular, an imperative occurrence of *περίμενε* does not establish third-person indicative past morphology. Plan a reviewed reusable lexical store with source/version, features, frame, backend, tests and review status; a successful parse alone must not promote a model hypothesis to reviewed vocabulary.

For deeper Svarna integration, the useful external change is an endpoint returning lemma, available morphological/dependency annotations, stable source IDs, corpus/variety and split/provenance information. Existing annotations and model predictions must be distinguished. This requires work on Svarna itself; the DS adapter cannot invent absent fields. Until source overlap can be excluded, disable retrieval from evaluation corpora when claiming held-out performance. Triantafyllidis can continue through bounded lookup; a bulk index would require separate data-use and maintenance decisions.

Priority 3: implement construction families in both native semantic systems. Select the first families from measured failures. Likely high-value Greek work includes plural/case agreement, productive finite paradigms, common complement clauses and *nov* relatives. Likely English work includes possessives and NP coordination, broader auxiliaries/aspect/passives, additional plural readings and *wh*-questions. Existing English polar questions, restricted relatives and distributive *all* remain useful starting points, not reasons to count those entire families as solved. Broaden PP frames and attachment explicitly: selected arguments, VP LINK modifiers and nominal attachment need distinguishable trees and meanings, not a rule making every PP an interchangeable argument.

Each family should supply reusable DS action templates, lexical applicability conditions, semantic expectations for Classical and Constructive output, positive examples, negative/ambiguous contrasts, and playback/export checks. Check Constructive exports with Coq where supported; inspect the assumptions as well as well-typedness. Keep tense/aspect promises tied to represented meanings, not merely to acceptance of more surface verb forms. A model may propose which implemented template fits; new rule families still need explicit implementation.

Priority 4: expand TTR on its own semantic foundations. Start with an audit of loaded legacy templates and semantic regression contracts: fresh individuals, subject/object/recipient roles, manifest constants, event/tense fields, scope, coreference, record dependencies and repair. Type-compatible but incorrectly merged records must fail these semantic checks. Retain the current collision tests while testing more compositional and dialogue cases.

The first expansion milestone is an English TTR lexical compiler for a stated set of existing, validated name/noun/verb frames. It can reuse source retrieval and model proposal infrastructure, but needs a TTR-specific schema contract, record builders and compatibility checks. Quantification, modifiers, complements and wider dialogue then require their own TTR derivations and tests. Greek TTR additionally requires Greek construction resources; native Greek coverage does not supply them automatically. Cross-backend comparisons should check agreed semantic relationships while respecting differences between choice terms, dependent witnesses and event records. Relabeling native output as TTR is not an implementation strategy.

Priority 5: improve paragraph interpretation and ambiguity. Expand anaphora beyond the current basic named antecedents, including common-noun discourse referents, agreement, accessible antecedents under negation/embedding and speaker changes. Retain alternative antecedents and PP attachments when the input does not choose between them. A paragraph may have complete individual sentences without a well-resolved discourse interpretation; report both. Extend dialogue questions, ellipsis and repair only with explicit context and binding conditions. The existing speaker-sensitive reflexive example is a small verified starting point, not general conversational understanding.

Priority 6: scale only after measuring the bottleneck, then release reproducibly. Profile alternative growth, completion/replay, source latency and model calls. Improve candidate deduplication and ordering while keeping alternatives available for recovery; document any pruning. The interactive Jev comparison now exposes single-sentence effects using identical candidates. Extend it to a frozen evaluation on the same grammar and data, measuring semantic quality as well as time under matched execution budgets and instrumentation. Enable an automatic preference only if the evidence supports a useful tradeoff. Simply increasing word limits or model calls would otherwise amplify latency and cost without necessarily improving coverage.

The current 55-second worker and 60-second request limits make larger jobs a separate hosting decision. If measurements justify it, add cancellable background jobs and suitable storage, or a separately hosted worker, while keeping the browser on Vercel. Persist public lexical evidence only with source conditions, cache invalidation and versioning; keep user text/key handling explicit. The next release should archive a tagged source revision, grammar and data hashes, dependency lock, evidence settings, provider/model identifiers and evaluation outputs. Staging manifests already provide part of this foundation.

What would count as progress. A milestone is complete when it improves a stated coverage family on fresh examples, satisfies reviewed semantic contracts, introduces no known overgeneration in the relevant contrast set, and stays within the reported resource budget. Publish counts and uncertainty rather than an unsupported headline percentage. Native English, native Greek, TTR and each dialect need separate coverage statements. Dialect expansion should follow source-reviewed rules and morphology for that variety, not automatic transfer from Standard Greek or its dictionary.

The immediate recommended work package is therefore: freeze the baseline, implement the morphology/lemma interface, and use the resulting failure counts to select the first productive construction family in each language. In the same development cycle, establish the next TTR semantic contracts and specify its first lexical compiler. The dated [wide-coverage plan](#) and [source integration design](#) remain useful implementation history; this section states the current forward priorities.

19 Community use, extension and maintenance

The community guide gives separate routes for teaching, reproducible experiments, linguistic evidence contributions, grammar development and maintenance. The README now starts from this checkout rather than implying that installing the upstream PyPI package reproduces the workbench.

The distribution includes the HTML/JS/SVG interface and selected runtime evidence in addition to the Python engine and grammar files. Resource lookup works both in editable source checkouts and installed wheels; writable caches remain in the source root or launch directory. Optional WordNet/Brown/Gutenberg indexes are separate. A clean-install check imports the wheel outside the checkout and exercises HTTP and streaming without model requests.

The executable contribution tutorial copies Classical and Constructive English grammars, adds a typed intransitive predicate, and checks twelve cases including negation, nonrestrictive relatives and failed contrasts. It writes file hashes and expected/observed meanings. Independent grammar folders load through the Python API; the public browser does not accept arbitrary grammar uploads. Use `top_n=0` when evaluating construction coverage: the inherited API default of three lexical alternatives can prune an entry needed by a function word.

The contribution guide, issue/PR templates and construction design template connect source evidence to DS operations and backend-specific semantic tests. The maintainer guide records process isolation, data conditions, hosting, release verification and remaining ownership decisions. A dedicated public repository, agreed contribution/licensing terms and a versioned source release remain necessary before claiming the extended project is fully handed over to the community. The licence mismatch is documented in section 2; the local packaging checks do not themselves resolve distribution rights.